

1	 Company Name Not Available Bedford, MA USA		Case Name: ANILINA_99.hsc				
2			Unit Set: SI10a1				
3							
4			Date/Time: Wed Jun 24 18:48:37 2026				
5							
6	Distillation: T-101 @Main						
7							
8							
9	CONNECTIONS						
10							
11	Inlet Stream						
12	Stream Name		Stage		From Unit Operation		
13	R-T101		Reboiler				
14	ORG-110		1_Main Tower		Heater		E-104
15	Outlet Stream						
16	Stream Name		Stage		To Unit Operation		
17	C-T101		Condenser				
18	H2O		Condenser		3 Phase Separator		V-102
19	ORG-SEP		Reboiler		Cooler		E-105
20	H2		Condenser				
21	MONITOR						
22							
23	Specifications Summary						
24		Specified Value		Current Value		Wt. Error	
25	Reflux Ratio	2.000 *		2.000		-2.839e-008	
26	Distillate Rate	---		14.36 kgmole/h		---	
27	Reflux Rate	30.16 kgmole/h *		28.85 kgmole/h		-4.341e-002	
28	Btms Prod Rate	---		41.62 kgmole/h		---	
29	Vent Rate	7.000e-002 kgmole/h *		7.000e-002 kgmole/h		3.773e-006	
30	REC H2O	0.9900 *		0.9900		9.044e-008	
31		Wt. Tol.		Abs. Tol.		Active	Estimate Used
32	Reflux Ratio	1.000e-002		1.000e-002		On	On On
33	Distillate Rate	1.000e-002		1.000 kgmole/h		Off	Off Off
34	Reflux Rate	1.000e-002		1.000 kgmole/h		Off	On Off
35	Btms Prod Rate	1.000e-002		1.000 kgmole/h		Off	On Off
36	Vent Rate	1.000e-002		1.000 kgmole/h		On	On On
37	REC H2O	1.000e-002		1.000e-003		On	On On
38	SPECS						
39							
40	Column Specification Parameters						
41	Reflux Ratio						
42							
43	Fix/Rang:	Fixed	Prim/Alter:	Primary	Lower Bnd:	---	Upper Bnd: ---
44	Stage:	Condenser	Flow Basis:	Molar	Liquid Spec:	---	
45	Distillate Rate						
46							
47	Fix/Rang:	Fixed	Prim/Alter:	Primary	Lower Bnd:	---	Upper Bnd: ---
48	Stream:	H2O	Flow Basis:	Molar			
49	Reflux Rate						
50							
51	Fix/Rang:	Fixed	Prim/Alter:	Primary	Lower Bnd:	---	Upper Bnd: ---
52	Stage:	Condenser	Flow Basis:	Molar	Liquid Spec:	---	
53	Btms Prod Rate						
54							
55	Fix/Rang:	Fixed	Prim/Alter:	Primary	Lower Bnd:	---	Upper Bnd: ---
56	Stream:	ORG-SEP	Flow Basis:	Molar			
57	Vent Rate						
58							
59	Fix/Rang:	Fixed	Prim/Alter:	Primary	Lower Bnd:	---	Upper Bnd: ---
60	Stream:	H2	Flow Basis:	Molar			
61							
62							
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8						
9	Column Specification Parameters					
10	REC H2O					
11						
12	Fix/Rang:	Fixed	Prim/Alter:	Primary	Lower Bnd:	---
13	Draw:	H2O	Flow Basis:	Molar		
14	Components:	H2O				
15	SUBCOOLING					
16						
17			Condenser			
18	Degrees of Subcooling				---	
19	Subcool to				---	
20	User Variables					
21						
22	ACTIVE INTERNAL OPTION: Internals-1@Main Tower@COL1					
23						
24	Name	Start Stage	End Stage	Mode	Internals	Tray / Packing Type
25	CS-1	1__ Main Tower	6__ Main Tower	Interactive Sizing	Trayed	Sieve
26						
27	Number of Passes	Packing Vendor	Packing Material	Packing Dimension	Tray Spacing / Section Packed Height (m)	Diameter (m)
28	1	---	---	---	0.6096	0.5044
29	SETUP					
30						
31	Section Name	CS-1				
32	Section Start	1__ Main Tower				
33	Section End	6__ Main Tower				
34	Internals	Trayed				
35	Internals Type	Sieve				
36	Diameter (m)	0.5044				
37	Tray Spacing / Section Packed Height (m)	0.6096				
38	Number Of Passes	1				
39	Maximum Acceptable Pressure Drop (kPa)	2.500				
40	Maximum Percent Downcomer Backup	100.00 %				
41	Maximum Percent Jet Flood	100.00 %				
42	Percent Jet Flood For Design	80.00 %				
43	Maximum Percent Liquid Entrainment	10.00 %				
44	Minimum Weir Loading (m3/h-m)	4.471				
45	Maximum Weir Loading (m3/h-m)	117.4				
46	Minimum Downcomer Area / Total Tray Area	0.1000				
47	Override Downcomer Froth Density	No				
48	Froth Density	---				
49	Weep Method	Hsieh				
50	Default Jet Flood Calculation Method	GLITSCH6				
51	Maximum Downcomer Loading Method	Glitsch				
52	% Approach to Maximum Capacity	---				
53	Design Capacity Factor	---				
54	Capacity Factor at Flooding	---				
55	System Foaming Factor	1.000				
56	Aeration Factor Multiplier	1.000				
57	Minimum Liquid Flow Rate	---				
58	Pressure Drop at Flood per Unit Packed Height	---				
59	Allowable Pressure Drop per Unit Packed Height	---				
60	Minimum Pressure Drop per Unit Packed Height	---				
61	Number of Curves	---				
62	Warning Status (% to Limit)	10.00 % *				
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8							
9	ACTIVE INTERNAL OPTION: Internals-1@Main Tower@COL1						
10							
11	SETUP						
12							
13	Pressure Drop Calculation Method		---				
14	Mode		Interactive Sizing				
15	Status		Needs Calculating				
16							
17	GEOMETRY DETAILS						
18	Common Geometry		CS-1				
19	Section Start		1__ Main Tower				
20	Section End		6__ Main Tower				
21	Internals		Sieve				
22	Section Diameter (m)		0.5044				
23	Foaming Factor		1.000				
24	Over-Design Factor		1.000				
25	Common Tray Geometry		CS-1				
26	Number of Passes		1				
27	Tray Spacing (m)		0.6096				
28	Picket Fence Weirs		No				
29	Swept Back Weirs		No				
30	Active Area Under Downcomer		No				
31	Deck Thickness		10 Gauge				
32	Deck Thickness Value (mm)		3.404				
33	Balance Downcomers Based On		Downcomer Loading				
34	Weir Modifications		None				
35	Net Area (m2)		0.1798				
36	Cross-Sectional Area (m2)		0.1998				
37	Active Area (m2)		0.1598				
38	Downcomer Geometry		CS-1				
39	Side	Weir Height (mm)	50.80 *				
40		Weir Length (m)	---				
41		Downcomer Clearance (mm)	38.10				
42		Downcomer Width - Top (mm)	78.92				
43		Downcomer Width - Bottom (mm)	78.92				
44		Downcomer Loading Top (m3/h-m2)	238.3 *				
45		Weir Loading (m3/h-m)	19.76				
46		Downcomer Area - Top (m2)	1.998e-002				
47		Downcomer Area - Bottom (m2)	1.998e-002				
48		Picketing Fraction	---				
49	Center	Weir Height (mm)	---				
50		Weir Length (m)	---				
51		Downcomer Clearance (mm)	---				
52		Downcomer Width - Top (mm)	---				
53		Downcomer Width - Bottom (mm)	---				
54		Downcomer Loading Top (m3/h-m2)	---				
55		Weir Loading (m3/h-m)	---				
56		Downcomer Area - Top (m2)	---				
57	Off Center	Downcomer Area - Bottom (m2)	---				
58		Picketing Fraction	---				
59		Weir Height (mm)	---				
60		Inside Weir Length (m)	---				
61		Outside Weir Length (m)	---				
62		Downcomer Clearance (mm)	---				
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9	ACTIVE INTERNAL OPTION: Internals-1@Main Tower@COL1						
10							
11	GEOMETRY DETAILS						
12							
13	Off Center	Downcomer Width - Top (mm)	---				
14		Downcomer Width - Bottom (mm)	---				
15		Downcomer Loading Top (m3/h-m2)	---				
16		Maximum Outside Weir Loading (m3/h-m)	---				
17		Maximum Inside Weir Loading (m3/h-m)	---				
18		Downcomer Area - Top (m2)	---				
19		Downcomer Area - Bottom (m2)	---				
20		Inside Picketing Fraction	---				
21		Outside Picketing Fraction	---				
22		Off-Center Downcomer Location (m)	---				
23	Swept Back Weir Geometry		CS-1				
24	Compatibility		KG Tower				
25	A		---				
26	B/Parallel Chord Segment		---				
27	S/Swept-Back Weir		---				
28	Swept-Back Weir Chord		---				
29	Angled Chord Segment		---				
30	Tray With Maximum Weir Loading		6				
31	Maximum Weir Loading (m3/h-m)		19.76 *				
32	Maximum Allowable Weir Loading in Section (m3/h-m)		117.4 *				
33	Actual Side Weir Length (m)		0.3665				
34	Effective Side Weir Length (m)		0.3665				
35	Lost Area (%)		0.00				
36	Sieve Geometry		CS-1				
37	Hole Diameter (mm)		12.70				
38	Number of Holes		126				
39	Hole Area to Active Area		0.1000				
40	Lattice Geometry		CS-1				
41	Number Of Downcomers		---				
42	Downcomer End Gap (mm)		---				
43	Downcomer Width (mm)		---				
44	Downcomer Height (m)		---				
45	Downcomer Spacing (m)		---				
46	Liq Exit Area / Bottom Downcomer Area		---				
47	Weir Height (mm)		---				
48	Bubble Cap Geometry		CS-1				
49	Cap Diameter		---				
50	Skirt Height		---				
51	Number of Caps		---				
52	Number of Caps Per Active Area		---				
53	Valve Geometry		CS-1				
54	Tray Type		---				
55	Valve Type		---				
56	Valve Material		---				
57	Leg Length		---				
58	Valve Thickness		---				
59	Number of Valves		---				
60	Number of Valves per Active Area		---				
61							
62							
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7	Distillation: T-101 @Main (continued)						
8							
9	ACTIVE INTERNAL OPTION: Internals-1@Main Tower@COL1						
10							
11	RESULTS SUMMARY						
12							
13	Section Name	CS-1					
14	Section Start	1_ Main Tower					
15	Section End	6_ Main Tower					
16	Internals	Trayed					
17	Diameter (m)	0.5044					
18	Number of Passes	1					
19	Tray Spacing / Section Packed Height (m)	0.6096					
20	Total Height (m)	3.658					
21	Total Pressure Drop (mbar)	40.89					
22	Total Pressure Drop (Head Loss) (mm)	437.5					
23	Trays With Weeping	None					
24	Maximum Percent Jet Flood (%)	80.00 *					
25	Tray With Maximum Jet Flood	6_ Main Tower					
26	Maximum Percent Downcomer Backup (%)	37.38 *					
27	Tray With Maximum Downcomer Backup	6_ Main Tower					
28	Maximum Downcomer Loading (m3/h-m2)	362.4 *					
29	Tray With Maximum Downcomer Loading	6_ Main Tower					
30	Maximum Downcomer Loading Location	Side					
31	Maximum Weir Loading (m3/h-m)	19.76 *					
32	Tray With Maximum Weir Loading	6_ Main Tower					
33	Maximum Weir Loading Location	Side					
34	Maximum Aerated Height Over Weir (mm)	104.1 *					
35	Tray With Maximum Aerated Height Over Weir	6_ Main Tower					
36	Maximum % Approach To System Limit (%)	50.90 *					
37	Tray With Maximum % Approach To System Limit	6_ Main Tower					
38	Maximum Cs Based On Bubbling Area (m/s)	0.1051 *					
39	Tray With Maximum Cs Based On Bubbling Area	6_ Main Tower					
40	Maximum % Capacity (Constant L/V)	80.00					
41	Maximum Capacity Factor (m/s)	---					
42	Section Pressure Drop (mbar)	40.89					
43	Average Pressure Drop Per Height (mbar/m)	---					
44	Average Pressure Drop Per Height (Friction) (mbar/m)	---					
45	Maximum Stage Liquid Holdup (m3)	---					
46	Maximum Liquid Superficial Velocity (m3/h-m2)	---					
47	Surface Area (m2/m3)	---					
48	Void Fraction	---					
49	1st Stichlmair Constant	---					
50	2nd Stichlmair Constant	---					
51	3rd Stichlmair Constant	---					
52	STAGE BY STAGE RESULTS: CS-1						
53							
54	State Conditions						
55	Stages	Liquid Temperature (C)	Vapor Temperature (C)	Liquid Mass Flow (kg/h)	Vapor Mass Flow (kg/h)	Liquid Volume Flow (m3/s)	Vapor Volume Flow (m3/s)
56							
57	1_ Main Tower	94.65	99.09	4592	860.9	1.323e-003	0.3389
58	2_ Main Tower	94.66	94.66	4592	673.9	1.323e-003	0.2702
59	3_ Main Tower	94.66	94.69	4592	674.1	1.323e-003	0.2702
60	4_ Main Tower	94.69	96.62	4601	683.0	1.325e-003	0.2711
61	5_ Main Tower	96.62	134.2	4961	1042	1.433e-003	0.2810
62	6_ Main Tower	134.2	179.3	6718	2800	2.011e-003	0.3360
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6	Distillation: T-101 @Main (continued)					
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8						
9	ACTIVE INTERNAL OPTION: Internals-1@Main Tower@COL1					
10						
11	STAGE BY STAGE RESULTS: CS-1					
12						
13	Stages	Volume (m3)	Residence Time (seconds)	Velocity From Top (m/s)	Velocity from Bottom (m/s)	Exit Velocity (m/s)
14	6 _Main Tower	3.000e-003	1.491	0.1007	0.1007	0.1440
15						
16	RATING					
17						
18	Tray Sections					
19						
20	Tray Section	Main Tower				
21	Tray Diameter (m)	1.500				
22	Weir Height (m)	5.000e-002		*		
23	Weir Length (m)	1.200		*		
24	Tray Space (m)	0.5500		*		
25	Tray Volume (m3)	0.9719				
26	Disable Heat Loss Calculations	No				
27	Heat Model	None				
28	Rating Calculations	No				
29	Tray Hold Up (m3)	8.836e-002				
30	Vessels					
31						
32	Vessel	Reboiler		Condenser		
33	Diameter (m)	1.193		1.193		
34	Length (m)	1.789		1.789		
35	Volume (m3)	2.000 *		2.000 *		
36	Orientation	Horizontal		Horizontal		
37	Vessel has a Boot	No		No		
38	Boot Diameter (m)	---		---		
39	Boot Length (m)	---		---		
40	Hold Up (m3)	1.000		1.000		
41	Other Equipment In Column Flowsheet					
42						
43						
44	Pressure Profile					
45						
46		Pressure (atm)		Pressure Drop (kPa)		
47	Condenser	1.000 atm *		0.0000 kPa		
48	1 _Main Tower	1.000 atm		0.0000 kPa		
49	2 _Main Tower	1.000 atm		0.0000 kPa		
50	3 _Main Tower	1.000 atm		0.0000 kPa		
51	4 _Main Tower	1.000 atm		0.0000 kPa		
52	5 _Main Tower	1.000 atm		0.0000 kPa		
53	6 _Main Tower	1.000 atm		---		
54	Reboiler	1.000 atm *		0.0000 kPa		
55	Pressure Solving Options					
56						
57	Pressure Tolerance	1.000e-004		Pressure Drop Tolerance		1.000e-004
58	Damping Factor	1.000		Max Press Iterations		100
59	CONDITIONS					
60						
61	Name	ORG-110 @Main	AN-NB FONDO @Main	H2-ANTORCHA @Main	H2O-NB @Main	
62	Vapour	0.1380	0.0000	0.9799	0.0000	
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7	Distillation: T-101 @Main (continued)					
8						
9						
10	CONDITIONS					
11	Temperature (C)	110.0000 *	179.3166	115.0786	95.0945	
12	Pressure (atm)	1.0000 *	1.0000	1.0000	1.0000	
13	Molar Flow (kgmole/h)	56.0413	41.6161	0.0700	14.3552	
14	Mass Flow (kg/h)	4219.9744	3918.2059	1.7046	300.0639	
15	Std Ideal Liq Vol Flow (m3/h)	4.0954	3.7954	0.0018	0.2982	
16	Molar Enthalpy (kJ/kgmole)	-2.567e+004	6.410e+004	-1.870e+005	-2.676e+005	
17	Molar Entropy (kJ/kgmole-C)	-9.918	4.819	144.8	21.16	
18	Heat Flow (kJ/h)	-1.4386e+06	2.6675e+06	-1.3091e+04	-3.8419e+06	
19	Name	R-T101 @Main	C-T101 @Main			
20	Vapour	---	---			
21	Temperature (C)	---	---			
22	Pressure (atm)	---	---			
23	Molar Flow (kgmole/h)	---	---			
24	Mass Flow (kg/h)	---	---			
25	Std Ideal Liq Vol Flow (m3/h)	---	---			
26	Molar Enthalpy (kJ/kgmole)	---	---			
27	Molar Entropy (kJ/kgmole-C)	---	---			
28	Heat Flow (kJ/h)	2.0260e+06	1.7749e+06			
29						
30	PROPERTIES					
31	Name	ORG-110 @Main	AN-NB FONDO @Main	H2-ANTORCHA @Main	H2O-NB @Main	
32	Molecular Weight	75.30	94.15	24.35	20.90	
33	Molar Density (kgmole/m3)	0.2265	9.392	3.203e-002	45.86	
34	Mass Density (kg/m3)	17.06	884.2	0.7801	958.7	
35	Act. Volume Flow (m3/h)	247.4	4.431	2.185	0.3130	
36	Mass Enthalpy (kJ/kg)	-340.9	680.8	-7680	-1.280e+004	
37	Mass Entropy (kJ/kg-C)	-0.1317	5.119e-002	5.944	1.012	
38	Heat Capacity (kJ/kgmole-C)	156.9	211.4	45.65	79.50	
39	Mass Heat Capacity (kJ/kg-C)	2.083	2.245	1.874	3.803	
40	LHV Molar Basis (Std) (kJ/kgmole)	2.423e+006	3.222e+006	3.003e+005	1.175e+005	
41	HHV Molar Basis (Std) (kJ/kgmole)	2.540e+006	3.364e+006	3.488e+005	1.621e+005	
42	HHV Mass Basis (Std) (kJ/kg)	3.373e+004	3.573e+004	1.432e+004	7755	
43	CO2 Loading	---	0.0000	---	0.0000	
44	CO2 Apparent Mole Conc. (kgmole/m3)	---	---	---	---	
45	CO2 Apparent Wt. Conc. (kgmol/kg)	---	---	---	---	
46	LHV Mass Basis (Std) (kJ/kg)	3.218e+004	3.422e+004	1.233e+004	5623	
47	Phase Fraction [Vol. Basis]	4.533e-002	0.0000	0.9264	0.0000	
48	Phase Fraction [Mass Basis]	4.437e-002	0.0000	0.9110	0.0000	
49	Phase Fraction [Act. Vol. Basis]	0.9828	0.0000	0.9999	0.0000	
50	Mass Exergy (kJ/kg)	-65.44	-107.5	232.9	6.246	
51	Partial Pressure of CO2 (atm)	0.0000	0.0000	0.0000	0.0000	
52	Cost Based on Flow (Cost/s)	0.0000	0.0000	0.0000	0.0000	
53	Act. Gas Flow (ACT_m3/h)	243.1	---	2.185	---	
54	Avg. Liq. Density (kgmole/m3)	13.68	10.96	38.49	48.14	
55	Specific Heat (kJ/kgmole-C)	156.9	211.4	45.65	79.50	
56	Std. Gas Flow (STD_m3/h)	1325	984.0	1.655	339.4	
57	Std. Ideal Liq. Mass Density (kg/m3)	1030	1032	937.3	1006	
58	Act. Liq. Flow (m3/s)	1.180e-003	1.231e-003	4.055e-008	8.694e-005	
59	Z Factor	---	2.868e-003	---	---	
60	Watson K	9.066	9.068	9.023	8.937	
61	User Property	---	---	---	---	
62	Partial Pressure of H2S (atm)	0.0000	0.0000	0.0000	0.0000	
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9	PROPERTIES					
10						
11	Name	ORG-110 @Main	AN-NB FONDO @Main	H2-ANTORCHA @Main	H2O-NB @Main	
12	Cp/(Cp - R)	1.056	1.041	1.223	1.117	
13	Cp/Cv	1.007	1.347	1.217	1.186	
14	Ideal Gas Cp/Cv	1.080	1.055	1.237	1.283	
15	Ideal Gas Cp (kJ/kgmole-C)	112.4	158.7	43.35	37.72	
16	Mass Ideal Gas Cp (kJ/kg-C)	1.493	1.685	1.780	1.805	
17	Heat of Vap. (kJ/kgmole)	9.329e+004	4.292e+004	6.539e+004	4.109e+004	
18	Kinematic Viscosity (cSt)	---	0.3813	---	0.4065	
19	Liq. Mass Density (Std. Cond) (kg/m3)	1034	1037	937.0	1028	
20	Liq. Vol. Flow (Std. Cond) (m3/h)	4.081	3.777	1.819e-003	0.2920	
21	Liquid Fraction	0.8620	1.000	2.014e-002	1.000	
22	Molar Volume (m3/kgmole)	4.414	0.1065	31.22	2.180e-002	
23	Mass Heat of Vap. (kJ/kg)	1239	455.9	2685	1966	
24	Phase Fraction [Molar Basis]	0.1380	0.0000	0.9799	0.0000	
25	Surface Tension (dyne/cm)	37.15	25.49	46.16	---	
26	Thermal Conductivity (W/m-K)	---	0.1159	---	0.5930	
27	Bubble Point Pressure (atm)	2.268	1.000	1750	0.9450	
28	Viscosity (cP)	---	0.3371	---	0.3897	
29	Cv (Semi-Ideal) (kJ/kgmole-C)	148.6	203.1	37.33	71.19	
30	Mass Cv (Semi-Ideal) (kJ/kg-C)	1.973	2.157	1.533	3.406	
31	Cv (kJ/kgmole-C)	155.7	157.0	37.50	67.06	
32	Mass Cv (kJ/kg-C)	2.068	1.667	1.540	3.208	
33	Cv (Ent. Method) (kJ/kgmole-C)	---	---	---	---	
34	Mass Cv (Ent. Method) (kJ/kg-C)	---	---	---	---	
35	Cp/Cv (Ent. Method)	---	---	---	---	
36	Reid VP at 37.8 C (atm)	1.333e-002	2.540e-003	---	5.019e-003	
37	True VP at 37.8 C (atm)	1.829	3.380e-003	2854	0.1235	
38	Liq. Vol. Flow - Sum(Std. Cond) (m3/h)	4.079	3.777	1.821e-003	0.2931	
39	Viscosity Index	-0.7113	-15.08	-35.46	-15.79	
40						
41	SUMMARY					
42	Flow Basis:	Molar	The composition option is selected			
43	Feed Composition					
44		ORG-110				
45	Flow Rate (kgmole/h)	56.0413				
46		---				
47	Hydrogen	0.0001				
48	NitroBZ	0.0302				
49	H2O	0.2493				
50	Aniline	0.7204				
51	Flow Basis:	Molar	The composition option is selected			
52	Feed Flows					
53		ORG-110				
54	Flow Rate (kgmole/h)	56.0413				
55		---				
56	Hydrogen (kgmole/h)	0.0066				
57	NitroBZ (kgmole/h)	1.6933				
58	H2O (kgmole/h)	13.9688				
59	Aniline (kgmole/h)	40.3727				
60	Products					
61	Flow Basis:	Molar	The composition option is selected			
62	Product Compositions					
63	Aspen Technology Inc.		Aspen HYSYS Version 14		Page 9 of 11	

1	 Company Name Not Available Bedford, MA USA				Case Name: ANILINA_99.hsc			
2					Unit Set: SI10a1			
3					Date/Time: Wed Jun 24 18:48:37 2026			
4								
5								
6								
7	Distillation: T-101 @Main (continued)							
8								
9	SUMMARY							
10								
11		H2		H2O		ORG-SEP		
12	Flow Rate (kgmole/h)	0.0700		14.3552		41.6161		
13		---		---		---		
14	Hydrogen	0.0945		0.0000		0.0000		
15	NitroBZ	0.0392		0.0045		0.0391		
16	H2O	0.8167		0.9633		0.0020		
17	Aniline	0.0496		0.0322		0.9589		
18	Flow Basis:	Molar		The composition option is selected				
19	Product Flows							
20		H2		H2O		ORG-SEP		
21	Flow Rate (kgmole/h)	0.0700		14.3552		41.6161		
22		---		---		---		
23	Hydrogen (kgmole/h)	0.0066		0.0000		0.0000		
24	NitroBZ (kgmole/h)	0.0027		0.0645		1.6260		
25	H2O (kgmole/h)	0.0572		13.8291		0.0825		
26	Aniline (kgmole/h)	0.0035		0.4617		39.9075		
27	Flow Basis:	Molar		The composition option is selected				
28	Product Recoveries							
29		H2		H2O		ORG-SEP		
30	Flow Rate (kgmole/h)	0.0700		14.3552		41.6161		
31		---		---		---		
32	Hydrogen (%)	99.6967		0.3033		0.0000		
33	NitroBZ (%)	0.1622		3.8070		96.0307		
34	H2O (%)	0.4093		99.0000		0.5907		
35	Aniline (%)	0.0086		1.1435		98.8479		
36	COLUMN PROFILES							
37								
38	Reflux Ratio:	2.000	Reboil Ratio:	0.7828	The Flows Option is Selected	Flow Basis:	Molar	
39	Column Profiles Flows							
40		Temp	Pres	Net Liq	Net Vap	Net Feed	Net Draws	
41		(C)	(atm)	(kgmole/h)	(kgmole/h)	(kgmole/h)	(kgmole/h)	
42	Condenser	95.1	1.000	28.85	---	---	14.43	
43	1 Main Tower	94.7	1.000	73.84	43.28	56.04	---	
44	2 Main Tower	94.7	1.000	73.84	32.23	---	---	
45	3 Main Tower	94.7	1.000	73.84	32.23	---	---	
46	4 Main Tower	94.7	1.000	73.78	32.23	---	---	
47	5 Main Tower	96.6	1.000	71.88	32.17	---	---	
48	6 Main Tower	134.2	1.000	74.19	30.26	---	---	
49	Reboiler	179.3	1.000	---	32.58	---	41.62	
50	Column Profiles Energy							
51		Temperature		Liq Enthalpy		Vap Enthalpy		
52		(C)		(kJ/kgmole)		(kJ/kgmole)		
53	Condenser	95.09		-2.676e+005		-1.870e+005		
54	1 Main Tower	94.65		-9.017e+004		-2.265e+005		
55	2 Main Tower	94.66		-9.017e+004		-2.265e+005		
56	3 Main Tower	94.66		-9.016e+004		-2.265e+005		
57	4 Main Tower	94.69		-8.939e+004		-2.265e+005		
58	5 Main Tower	96.62		-6.038e+004		-2.250e+005		
59	6 Main Tower	134.2		3.933e+004		-1.646e+005		
60	Reboiler	179.3		6.410e+004		6.989e+004		
61								
62								
63	Aspen Technology Inc.		Aspen HYSYS Version 14				Page 10 of 11	

1	 Company Name Not Available Bedford, MA USA				Case Name: ANILINA_99.hsc			
2					Unit Set: SI10a1			
3					Date/Time: Wed Jun 24 18:48:37 2026			
4								
5								
6	Distillation: T-101 @Main (continued)							
7								
8								
9	FEEDS / PRODUCTS							
10								
11	Flow Basis:			Molar				
12		Stream	Type	Duty (kJ/h)	Phase	Flows (kgmole/h)	Enthalpy (kJ/kgmole)	Temp (C)
13								
14	Condenser	C-T101	Energy	1.8e+006 *		---	---	---
15		H2	Draw	---	Vapour	7.00e-002 *	-1.9e+005 *	95.09 *
16		H2O	Draw	---	Liquid	14.4 *	-2.7e+005 *	95.09 *
17	1__Main Tower	ORG-110	Feed	---	Mixed	56.0 *	-2.6e+004 *	110.00 *
18	2__Main Tower							
19	3__Main Tower							
20	4__Main Tower							
21	5__Main Tower							
22	6__Main Tower							
23	Reboiler	R-T101	Energy	2.0e+006 *		---	---	---
24		ORG-SEP	Draw	---	Liquid	41.6 *	6.4e+004 *	179.32 *
25	PERFORMANCE SUMMARY FOR INTERNAL OPTION: Internals-1@Main Tower@COL1							
26								
27	Number Of Stages				6 *			
28	Total Height (m)				3.658			
29	Total Head Loss (mm)				437.5			
30	Total Pressure Drop (mbar)				40.89			
31	Number Of Sections				1			
32	Number Of Diameters				1			
33	Pressure Drop Across Sump (kPa)				---			
34	Section	Start	End	Height (m)	Diameter (m)			
35								
36	CS-1	1__Main Tower	6__Main Tower	0.6096	0.5044			
37	Internals Type	Tray or Packing Type	Section Pressure Drop (mbar)	Approach To Flood (%)	Limiting Stage			
38								
39	Trayed	Sieve	40.89	80.00				
40								
41								
42								
43								
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63	Aspen Technology Inc.		Aspen HYSYS Version 14			Page 11 of 11		